e Andrew Jonathan Balmer

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Linkedin | Github

Microbial Genomics | Computational Biology · Open-Source Methods Development · Bioinformatics

Education

2018-2023: University of Cambridge: DPhil (PhD) Biological Sciences.

- Awarded prestigious BBSRC DTP Studentship Award (£70k, 4.5 years).

2014-2016: University of Cambridge: MPhil (Masters) Biological Sciences.

- Published first-author paper based on work from M. Phil. thesis.

2010-2014: University of Leeds: BSc Biology (International)

- 1st Class Honours degree; graduated top 5 in cohort.
- Study abroad: University of Heidelberg, Germany (Evolution and Genetics)

Research Experience

2023-2025 Senior Genomic Surveillance Data Scientist, Wellcome Sanger Institute

- Led meta-analysis of 100k+ *Plasmodium falciparum* genomes to investigate artemisinin resistance.
- Analysed population-scale microbial datasets, including heterogeneous samples, integrating epidemiological metadata to study pathogen dynamics over time and across space.
- Developed and adapted computational methods and scalable pipelines for handling complex microbial sequencing data—skills directly transferable to metagenomic workflows.
- Co-led development of PepExplorer, a novel peptide vaccine discovery platform for *P. falciparum*.
- Co-led grant development/research strategy with funders and international collaborators.

2018-2023 D. Phil. (PhD) Biological Sciences, University of Cambridge

- Developed 'Antimicrobial Resistance Cartography'—a computational framework for modelling multivariate drug resistance and diversity patterns across microbial populations, integrating dimensionality reduction, imputation, and genotype–phenotype mapping.
- Published in *PLOS Genetics*, *PNAS* and *BioRxiv*; presented at national & international conferences.

2022 Research Internship, UK Health Security Agency (UKHSA)

- Analysed emergence of SARS-CoV-2 mutations following antiviral treatment.
- Contributed to real-time surveillance of therapeutic resistance in COVID-19 variants.

2017-2018 Research Internships, University of Cambridge

- Conducted 200-day experimental evolution project, generating 291 mutant *Streptococcus suis* lines.
- Led meta-analysis of >100 studies on pathogen virulence evolution.
- Applied structural equation modelling (OpenMx) to influenza vaccine response data.

Project Management Experience

2016-2018: Synthetic Biology Strategic Research Initiative Assistant Coordinator

- Grew an online research community to 2,000+ followers through strategic content management.
- Organised international conference with 100+ attendees and keynote speakers from two countries.

2012-2014: Founder and President of LUU Science magazine

- Founded and scaled student-led popular science publication into an established magazine.
- Managed editorial team of 3; securing funding and advertising partnerships
- Awarded University of Leeds Partnership Award for Innovation

Research Skills

Programming and Data Science:

- R (5+ years), Python (2+ years), Shell scripting, version control/reproducibility (GitHub, Jupyter).
- Advanced statistical modeling: Genotype-phenotype association, multivariate analysis, MDS, machine learning, model fitting, data visualisation and reproducible research workflows.
- HPC/Cloud computing: Sanger cluster, LSF scheduling, Google Colab, parallel computing.
- Familiarity with challenges in metagenomics (community diversity, taxonomic profiling, mixed-sample interpretation) and experience analysing heterogeneous microbial datasets at scale.

Project management and leadership:

- Coordinated multi-partner projects on malaria genomics, managing timelines and deliverables across the Wellcome Sanger Institute, UNSW, and Stanford collaborators.
- Contributed to grant applications and progress reports; supported cross-functional teams (bioinformatics, immunology, epidemiology) to achieve research objectives.
- Designed and led large-scale lab experiments and data analysis pipelines.

Science communication:

- Strong presenter for academic audiences; delivered invited talks at CDC, UKHSA, and conferences.
- Published popular science articles in Scientific American and Yahoo News.

Laboratory skills:

- Experimental evolution, microbial culture, growth curves, DNA extraction, and PCR.

Languages: English (Native), German (CEFR-B2 - intermediate), Spanish (CEFR-A1, beginner).

Awards and Funding

- **BBSRC DTP Studentship Award**, 2018-2024 (£70,000)
- **Masters Bursary (Science) 2014,** 'Postgraduate Search' (£1000)
- **University of Leeds Partnership Award for innovation**, 2013 (LUU Science magazine).
- **Santander Research Scholarship (2012) -** For research into developmental genetics of *Arabidopsis thaliana* at the University of Heidelberg (£500)

Key Publications

- **Balmer AJ**, et al., (2025). Understanding the Global Rise of Artemisinin Resistance: Insights from over 100k *Plasmodium falciparum* Samples. eLife
- **Balmer AJ**, et al., (2025). Antimicrobial Resistance Cartography: A Generalisable Framework for the Study of Multivariate Drug Resistance. BiorXiv.
- Brizuela J, Murray GGR, Boueroy P, **Balmer AJ** (2025). Emergence of zoonotic and multi-drug resistant *Streptococcus suis.* BiorXiv.
- **MalariaGEN (2025).** Pf8: an open dataset of *Plasmodium falciparum* genome variation in 33,325 worldwide samples. Wellcome Open Research.
- Murray GGR, Hossain ASMM, Miller EL, Bruchmann S, **Balmer AJ**, *et al.,* (2023) The emergence and diversification of a zoonotic pathogen from within the microbiota of intensively farmed pigs. PNAS.
- Murray GGR, **Balmer AJ**, *et al.* (2021). 'Mutation rate dynamics reflect ecological changes in an emerging zoonotic pathogen'. PLOS Genetics.

Invited Talks

- **2024 – MalariaGEN Journal club.** 'Understanding the Global Spread of Artemisinin Resistance'.
- **2023 – Invited talks at UKHSA Therapeutics Group**, **CDC and Wellcome Sanger Institute**
    - 'Multivariate Methods for the Study of Multivariate Drug Resistance in Streptococci'.

Public Outreach

- GEM Conference Poster and Flash talk (2024).
- Organised and presented a series of lectures and podcasts on science communication (2013).

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